Cloud SQL - Michaël Bettan

Definition: Fully-managed relational database service supporting MySQL, PostgreSQL and MS SQL Server. Designed for relational data (record-based): tables, rows and columns, and super structured data. SQL compatible and can update fields.	**Availability**: Multiple zones Single zone	**SLA**: 99.95%

Use Cases: SQL DB scaling vertically for structured data
- OLTP workloads (OnLine Transaction Processing)
- Lift & Shift of legacy SQL RDBMS
- Web Frameworks, ERP, CRM, SaaS Applications
- Back-ends DB applications
- Ecommerce and websites (e.g., blogs)

Billing:
- Instance
- CPU
- Memory
- Storage
- Network Cost
- Licensing Cost (SQL Server only)

Key capabilities

- ***Automatic encryption***
- *Up to **64TB storage** capacity*
- *Up to **60,000 IOPS (read/write per second)***
- ***Auto-scale** (vertically) and **auto-backup***
- *Up to **96 cores** and **624 GB ram***
- ***Read replica** (read only) to offload work from an instance. The read replica is an exact copy of the primary instance. Data and other changes on the primary instance are updated in almost real time on the read replica.*
- ***Cascading replication**: create a read replica under another read replica in the same or a different region:*
- ***Disaster recovery:** You can use a cascading hierarchy of read replicas to simulate the topology of your primary instance and its read replicas. During an outage, your selected read replica is promoted to primary and the read replicas under the new primary continue to replicate and are ready for use.*
- ***Performance improvements:** Reduce the burden on the primary instance by offloading replication work to multiple read replicas.*
- ***Scale Reads:** You can have more replicas to share the read load.*
- ***Cost reduction:** You can reduce networking costs by using a single cascading replica with cross-region replication in other regions.*

Operational Model

- **Google's owns following responsibilities:**
  - IaaS Compute (hardware, virtualization)
  - OS install, upgrade, patches, ...
  - DB install, upgrade, patches, ...
  - Database Backups
  - High-Availability (failover)
  - Scaling (disk space)
  - Monitoring
  - Network connectivity
- **MySQL versions:** 8.4, 8.0, 5.7, 5.6
- **PostgreSQL versions:** 16, 15, 14, 13, 12, 11, 10, 9.6
- **SQL Server versions**: 2022, 2019, 2017

Migration Steps

- *Providing details on your data source, create a read replica, sync read replica with source and promoting read replica to primary instance*

Consideration

- *Critical for **managing new transactions***
- *Optimized for **high throughput WRITES** to **RECORDS**.*
- *Cloud SQL is **record-based** → entire record must be opened on disk even if you SELECTed only a single column*
  - *Cloud SQL **record-based** vs BQ **column-based***
- *Cloud SQL optimized for **transactions (writes)** whereas BigQuery for **reporting workloads (mostly reads)***
- *Limiting factors → Global availability & Disk size*
- *Connectivity with App Engine, GKE, GCE, Cloud Run*
- *Automate Backups (window of time) and enable point-in-time recovery (via write-ahead log archiving)*
- *Read replicas limited to the same region as the master*
- ***SSD** over HDD (lower performance)*
- ***IOPS** and **Network throughput** depends on the machine type and the configuration (bigger the faster)*
- ***Federated Query** from BigQuery*
- *Cloud Storage acts as the staging ground to move data between Cloud SQL and BigQuery.*
- *No autoscaling for the compute (nodes) supported*

Best Practices

- *Move smaller table rather than large tables*
- *Private IP → Private services access connection required*
- *Cloud SQL Auth Proxy → no authorized network, nor SSL*
- *Database failure → exponential backoff*
- *Enable **automatic storage increases***

Loading & Exporting the data

- Storage Transfer Service (STS) or Transfer Appliance to load data into Cloud Storage (from AWS, local, GCS)
- Use gsutil if copying files over from on-premise
- Cloud Storage as temporary staging
- Import/Export SQL dump/CSV file format
- Consider compressed gzip for saving cost

Security Access Control

- *Secret & Service acct to connect to SQL Instance from GKE*

Monitoring & Logging

- ***Cloud SQL Insights*** *7-day retention → detect, diagnose, and identify the root cause of performance problems*
- ***MySQL connector*** *required for* ***Stackdriver*** *monitoring*

Miscellaneous

- ***RPO****: Recovery Point Objective*
- ***RTO****: Recovery Time Objective*
- ***Database Migration*** *creates Migration jobs that allow you to move data between source and destination databases. You can define your source by creating a new job, or just its connection profile*

Auth Proxy

- *The Cloud SQL Auth proxy offers enhanced security, especially for applications with dynamic IP addresses. It eliminates the need to manage Authorized Networks or allowlisting IPs, simplifying connection management.*
- *It establishes secure connections by creating a secure tunnel. Database traffic is encapsulated and forwarded through this tunnel, leveraging Google IAM for authentication.*
- *By leaving Authorized Networks empty, you enforce exclusive use of the Auth proxy. This prevents any direct IP-based connections to your Cloud SQL instance, maximizing security.*